

Activity

Name: R. Rohul

reg NO: 192521362

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2) $\text{push}(54); \text{push}(52); \text{pop}(); \text{push}(55); \text{push}(62); S = \text{pop}();$
 $(54, 52) \quad (52) \quad (54, 55) \quad (54, 55, 62) \quad (62)$

$S = 62$

$\text{enqueue}(21); \text{enqueue}(24); \text{dequeue}(); \text{enqueue}(22); \text{enqueue}(32);$

$q = \text{dequeue}();$
 $(21, 24) \quad (21) \quad (24, 22) \quad (24, 22, 32)$

The

$(24) \quad q = 24$

$S + q = 62 + 24$
 $= 86$

3) the result evaluating postfix expression $10 \ 5 + 60 \ 6 / * 8 -$ is

$10 \ 5 + = 15$
 $60 \ 6 / = 10$
 $15 * 10 = 150$
 $150 - 8 = 142$

3) $8, 2, 3 \wedge / 2. 3 * + 5 1 * - 1$

Stack after first * evaluated;

Symbol	Stack
8	8
2	8, 2
3	8, 2, 3
$\wedge$	8, 8, $(2^3 = 8)$
/	1 $[8/8 = 1]$
2	1, 2
3	1, 2, 3
*	1, 6 $[2 * 3 = 6]$

The top two elements are 1, 6

4. $a + b * c - d ^ e f$

character	stack	o/p
a		a
+	+	ab
b	+	ab
*	+	abc
*	+	abc*
c	-	abc*+d
-	-	abc*+d:
d	-	abc*+de
^	-	abc**+def
e	-	abc*+def
f	-	abc*+def
empty		

5. Stack operations:-

	Stack	o/p
push(10)	10	-
push(10, 20)	10, 20	-
pop()	10	20
push(10)	10, 10	-
push(20)	10, 10, 20	-
pop	10, 10	20
pop	10	10
pop	empty	10
push(20)	20	-
pop	empty	20

Sequence of the popped values $\rightarrow 20, 20, 10, 10, 20$